

Rishi D. Jha

PHD STUDENT · MACHINE LEARNING

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Education

Cornell Tech

PHD STUDENT, COMPUTER SCIENCE

- Advisor: Prof. Vitaly Shmatikov

New York / Ithaca, NY

Aug. 2023 - Present

University of Washington — Seattle

MS., COMPUTER SCIENCE

- Master's Thesis: *Label Poisoning is All You Need*
- Advisor: Prof. Sewoong Oh

Seattle, WA

Sep. 2022 - Jun. 2023

University of Washington — Seattle

BS.BA., COMPUTER SCIENCE AND MATHEMATICS — PHILOSOPHY: *Cum Laude, Phi Beta Kappa*

- Jun. 2022: Graduated Cum Laude with Phi Beta Kappa honors
- 2018-22: Dean's List (all eligible quarters)

Seattle, WA

Sep. 2018 - Mar. 2022

Awards and Honors

2025	Doctoral Fellow , Digital Life Initiative	New York, NY
2024	Distinguished Paper Award , USENIX Security — Top 22 papers (out of 417)	Philadelphia, PA
2024	Graduate Research Fellowship Program Honorable Mention , NSF	USA
2023	Cornell University Fellow , Cornell University — 20% of incoming PhDs	Ithaca, NY
2022	Phi Beta Kappa , University of Washington	Seattle, WA
2022	Cum Laude , University of Washington — Top 10% across Arts & Sciences	Seattle, WA
2018-22	Dean's List , University of Washington — All eligible quarters	Seattle, WA
2021-22	Varsity Climbing Team , University of Washington	Seattle, WA
2019	Finalist , (Top 4 of 36 Teams) UW Foster CBDC: Consulting Challenge	Seattle, WA

Publications

CONFERENCE

- [1] **Rishi Jha**, Collin Zhang, Vitaly Shmatikov, and John X. Morris. "Harnessing the Universal Geometry of Embeddings". In: *Thirty-ninth Conference on Neural Information Processing Systems (NeurIPS)*. Dec. 2025.
- [2] Harold Triedman, **Rishi Jha**, and Vitaly Shmatikov. "Multi-Agent Systems Execute Arbitrary Malicious Code". In: 2nd Conference on Language Modeling (**COLM**). Mar. 2025.
- [3] Tingwei Zhang*, **Rishi Jha***, Eugene Bagdasaryan, and Vitaly Shmatikov. "Adversarial Illusions in Multi-Modal Embeddings". In: *33rd USENIX Security Symposium (USENIX)*. Received the **Distinguished Paper Award** (5% of accepted papers). Aug. 2024.
- [4] **Rishi Jha***, Jonathan Hayase*, and Sewoong Oh. "Label Poisoning is All You Need". In: *Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS)*. Dec. 2023.
- [5] Dimitrios C. Gklezakos, **Rishi Jha**, and Rajesh P.N. Rao. "Hyper-Universal Policy Approximation: Learning to Generate Actions from a Single Image using Hypernets". In: *Neurovision 2022: A CVPR Workshop (Neurovision @ CVPR)*. June 2022.
- [6] **Rishi Jha** and Kai Mihata. "On Geodesic Distances and Contextual Embedding Compression for Text Classification". In: *Proceedings of the Fifteenth Workshop on Graph-Based Methods for Natural Language Processing (TextGraphs-15 @ NAACL)*. June 2021.

MASTER'S THESIS

- [7] **Rishi Jha**. "Label Poisoning is All You Need". University of Washington, Seattle, 2023.

PATENTS

- [8] Nisha S. Hameed, **Rishi D. Jha**, and Evan Argyle. "Graph-Based Analysis of Security Incidents". U.S. pat. 12081569. Microsoft Technology Licensing LLC. Sept. 3, 2024.

PREPRINTS

- [9] **Rishi Jha**, Harold Triedman, Justin Wagle, and Vitaly Shmatikov. "Breaking and Fixing Defenses Against Control-Flow Hijacking in Multi-Agent Systems". Submitted to the Fourteenth International Conference on Learning Representations (**ICLR**). Sept. 2025.
- [10] Tingwei Zhang, Fnu Suya, **Rishi Jha**, Collin Zhang, and Vitaly Shmatikov. "Adversarial Hubness in Multi-Modal Retrieval". Submitted to the 46th ACM Conference on Computer and Communications Security (**CCS**). Apr. 2025.

Invited Talks & Guest Lectures

INVITED

2025	Tufts , (Dec.) “Harnessing the Universal Geometry of Embeddings.” <i>Computational Platonic Space</i> .	Remote
2025	UC Berkeley , (Nov.) “Breaking and Fixing Multi-Agent Systems.” <i>Security & AI Seminar</i> .	Berkeley, CA
2025	Microsoft ASG , “Contextual Control and Information Flow in Multi-Agent Systems.”	Redmond, WA
2025	Planet Labs , “Harnessing the Universal Geometry of Embeddings.”	Remote
2025	MSR , “Multi-Agent Systems Execute Arbitrary Malicious Code.” <i>Security & Privacy Workshop</i> .	Redmond, WA
2025	MongoDB , “Multi-Agent Systems Execute Arbitrary Malicious Code.”	New York, NY
2025	Microsoft ASG , “Emergent Risks in Multimodality.”	Remote

GUEST LECTURES

2025	Cornell Tech , “Security & Privacy in Machine Learning.”	New York, NY
2021	The Overlake School , “Topics on Sparse Coding.”	Remote

Academic Research

Cornell Security

New York, NY

PHD STUDENT

Aug 2023 – Present

Working with **Prof. Vitaly Shmatikov** understanding how generative AI’s internal knowledge representations—such as embeddings and inter-agent communications—create privacy and security vulnerabilities.

- Introduced **vec2vec**, the first unsupervised method for unpaired cross-model embedding translation—showing encoders share a *universal geometry* across paradigms and vintages; enables cross-model interpretability and reveals sensitive-data leakage risks in vector databases. **NeurIPS 2025** [1].
- Discovered *control-flow hijacking* attacks in LLM-based multi-agent systems that exploit inter-agent communication to trigger arbitrary code execution and data exfiltration, achieving 45–100% success rate. **CoLM 2025** [2].
- Demonstrated *adversarial illusions* in commercial multimodal embeddings—small perturbations align an image/audio embedding to an arbitrary cross-modal target, enabling task-agnostic attacks. **Distinguished Paper Award, USENIX Security 2024** [3].
- Introduced *adversarial hubs* enabling universal content injection and targeted attacks in multimodal retrieval. In submission [10].

Sewoong Lab — Foundations of Machine Learning

Seattle, WA

GRADUATE RESEARCH ASSISTANT

May 2021 – Aug. 2023

Worked with **Prof. Sewoong Oh** and **Jonathan Hayase** on backdoor attacks and representation learning.

- Developed FLIP, a trajectory-matching backdoor attack that poisons only the training labels—leaving inputs untouched—to inject arbitrary triggers; achieves 99.6% success with <1% label corruption and <1% clean accuracy drop. Robust across datasets, triggers, and architectures. **NeurIPS 2023**, Master’s thesis [4, 7].
- Created an open-source backdoor attack benchmark suite and survey: **backdoor-suite**.

Center for Neurotechnology

Seattle, WA

UNDERGRADUATE ML RESEARCHER

Mar. 2020 – Aug. 2022

Worked with **Prof. Rajesh Rao** and **Dimitrios Gklezakos** on hypernetworks and neural representation learning.

- Co-developed HUPA, a personalized hypernetwork for task-conditional RL with up to 35% more resilience to sparsity and 25% better generalization than traditional embeddings. **NeuroVision @ CVPR 2022** [5].
- Built an audio-visual hypernetwork where video inputs control audio processing weights for large-scale cross-modal learning.
- Designed a convolutional manifold-learning network for unsupervised sparse feature learning in natural images.

Other Research

Seattle, WA

UNDERGRADUATE STUDENT

Nov. 2018 – Jun. 2021

- (Self-Directed, Nov. 2020 – Jun. 2021) Worked with **Kai Mihata** to show that nonlinear compression of BERT embeddings can preserve downstream performance, making it practical for memory-constrained settings. **TextGraphs @ NAACL 2021** [6].
- (ICTD Lab, Nov. 2018 – May 2019) Worked with **Spencer Sevilla** to evaluate chat app performance under poor network conditions and implement a teaching solution for schoolchildren in rural Indonesia.

Industry Research

Microsoft

Redmond, WA

RESEARCH INTERN

May. 2025 – Aug. 2025

Worked with **Justin Wagle** and **Kazuhito Koishida** on contextual multi-agent defenses.

- Demonstrated control-flow hijacking attacks evade alignment checks and goal drift detection in multi-agent systems due to brittle definitions and incomplete context visibility. Designed and implemented **ControlValve**, applying control-flow integrity and least privilege to automatically generate and enforce permitted control-flow graphs with zero-shot contextual rules. In submission [9].
- Project chosen as the core safety layer for upcoming deployments; internal conversations kicked off cross-team planning.

Microsoft Defender Research

Redmond, WA

SOFTWARE ENGINEERING INTERN — DATA SCIENCE

Jul. 2022 - Sep. 2022

- Ideated, pitched, and implemented a low-cost, humanly interpretable meta-learning framework that exploits spectral similarities in existing classifier responses to drive robustness in the Defender product. The productionalized system was lightweight, had upwards of 97% precision and recall, and was humanly interpretable.
- After my departure, the model was being pushed from pre-production to production with the goal of providing protection for billions of users by the **end of 2023**.

Microsoft Defender Research

Remote

SOFTWARE ENGINEERING INTERN — DATA SCIENCE

Jun. 2021 - Sep. 2021

Patented in **September 2024** [8].

- Ideated and designed patented approach to detect malicious Command-and-Control intrusions in corporate networks using spectral methods on graphs. The model achieved high precision and recall in finding Indicators of Compromise in historical data.
- The project received significant investment from the team and Microsoft Research (MSR) since my departure with a goal of pushing an extension of the model to production in **Summer 2023**.

Teaching

Cornell Tech

New York, NY

TA FOR "TRUSTWORTHY AI"

Aug. 2025 - Dec. 2025

TA for the graduate-level trustworthy AI course with **Prof. Vitaly Shmatikov** and **Prof. Jack Morris**.

Cornell Tech

New York, NY

TA FOR "SECURITY AND PRIVACY CONCEPTS IN THE WILD"

Jan. 2025 - May. 2025

TA for the graduate-level security and privacy course with **Prof. Vitaly Shmatikov**.

Cornell Tech

New York, NY

TA FOR "PRIVACY IN THE DIGITAL AGE"

Aug. 2024 - Dec. 2024

TA for the graduate-level privacy course with **Profs. Vitaly Shmatikov** and **Helen Nissenbaum**.

University of Washington — Seattle

Seattle, WA

4X UNDERGRAD / GRAD MACHINE LEARNING TA

Mar. 2020 - Dec. 2021

During Spring 2020, Winter 2021, Spring 2021, Autumn 2021:

- Taught undergraduate and graduate students as an undergraduate through 25-person sections and biweekly office hours.
- Designed section materials for entire teaching staff, monitored discussion boards, and graded assignments.

University of Washington — Seattle

Seattle, WA

MACHINE LEARNING COURSE DESIGNER

Jun. 2021 - Sep. 2021

Redesigned course materials and infrastructure to improve accessibility and equity in ML education, under **Prof. Sewoong Oh**.

- Revamped problem sets and homework infrastructure to reflect a fast-evolving field and lower barriers to entry.
- Incorporated data context and fairness considerations to help students critically engage with algorithmic bias.
- Developed a centralized grading system and TA codebase for long-term course support.

Service

2025 **Host**, NYC Privacy Day

New York, NY

2025 **Strategic Officer**, PhDs at Cornell Tech (PACT)

New York, NY

2025 **Reviewer**, ICLR

Remote

2025 **Reviewer**, NeurIPS

Remote

2025 **Reviewer**, COLM

Remote

2024 **Reviewer**, ICLR

Remote

2023 **Reviewer**, ICLR

Remote

2023 **Reviewer**, ICML

Remote

Selected Coursework

Machine Learning	Machine Learning [†] , NLP [†] , Deep Learning Theory [†] , Reinforcement Learning [†] , Deep Learning
Other Computer Science	Cryptography [†] , Privacy [†] , Security [†] , Human-Centered AI [†] , Algorithms, Databases
Mathematics	Real Analysis I & II, Probability and Statistics I, II, & III, Modern Algebra I & II, Linear Algebra

[†]Taken at both the undergraduate and PhD levels.

[†]Taken at the PhD level.